Supply Brain — SKU Segmentation

Feature Engineering & Clustering Attribute Reference

This document lists every attribute used in building the SKU feature matrix, the raw data file each one is derived from, a plain-language explanation, whether it was used as an input to the k-means clustering itself, and — for the attributes that were used in clustering — an importance ranking based on an ANOVA F-test (a statistical test of how strongly each feature actually differentiates the resulting clusters).

1. Full Attribute Reference

Attribute	Source File	Explanation	Used in Clustering?	Importance (if used)
sku id	sku_master.csv	Unique identifier for each product.	No — identifier	—
category	sku_master.csv	Product category (Beverages, Dairy & Bakery, Frozen Foods, Grocery, Snacks & Confectionery). Used to look up category-level elasticity and cost parameters, not fed into clustering directly.	No	—
list price usd	sku_master.csv	Standard catalog price per unit.	Yes	Rank 4 (F=260)
unit cost usd	sku_master.csv	Cost to produce/source one unit. Used to derive margin_pct and realized margin; not itself a clustering input.	No — intermediate	—
margin pct	sku_master.csv	Profit margin as a % of list price: (price − cost) / price.	Yes	Rank 7 (F=3, borderline significant)
shelf life days	sku_master.csv	Number of days before the product expires.	Yes	Not statistically significant
segment	sku_master.csv	Pre-existing label (Fast-Moving / Intermediate / Seasonal), set before this analysis. Deliberately excluded from clustering — used only afterward to sanity-check our own clusters, to avoid circular reasoning.	No — deliberately excluded	—
seasonal flag	sku_master.csv	Pre-existing True/False seasonal tag. Used only in the separate time-cycle diagnostic (to validate seasonality), not in clustering.	No	—
demand cv	demand_forecast.csv	Demand volatility: standard deviation of weekly actual sales ÷ mean weekly sales, computed across 2025's real transaction history.	Yes	Rank 6 (F=56)
forecast mape	demand_forecast.csv	Average forecast error, ignoring direction (Mean Absolute Percentage Error) — how far off weekly forecasts typically run.	Yes	Not statistically significant
elasticity base	orders.csv (pooled per category, SKU+week fixed-effects regression)	Sensitivity of demand to everyday (non-promotional) price changes.	No — deliberately excluded	—

Attribute	Source File	Explanation	Used in Clustering?	Importance (if used)
elasticity promo	orders.csv (pooled per category)	Sensitivity of demand to promotional discounts.	No — deliberately excluded (only 5 distinct values across all 500 SKUs; would dominate and distort distance calculations)	—
revenue total	orders.csv	Total realized revenue from actual transactions across 2025 (unit price actually paid × quantity, summed).	Yes	Rank 5 (F=79)
total margin usd realized	orders.csv	Total realized profit in dollars: (unit price paid – unit cost) × quantity, summed across all 2025 transactions.	No — used for business-dollar profiling, not clustering input	—
supply adherence std	production.csv	Variability in how consistently production hit its planned output for that SKU.	Yes	Not statistically significant
lead time mean days	network_lanes.csv + plant_dc_master.csv (full plant→PDC→RDC→CDC path traced per SKU)	Average replenishment time from plant to the SKU's actual selling location(s), demand-weighted across DCs.	Yes	Rank 3 (F=266)
lead time std days	network_lanes.csv + plant_dc_master.csv	Unpredictability of that replenishment time.	Yes	Rank 2 (F=278)
stockout rate	inventory.csv	% of tracked weeks the SKU was completely out of stock (closing stock = 0).	Yes	Rank 1 — most dominant (F=354)
stockout cost multiplier	cost_parameters.csv (category-level lookup)	A risk-weighting multiplier used only to calculate financial exposure ($), not fed into the clustering itself.	No	—
stockout exposure usd	Derived: revenue_total × stockout_rate × stockout_cost_multiplier	Estimated dollar value at risk from stockouts. A profiling/reporting metric calculated AFTER clusters are formed.	No	—

Note: 10 of the 20 attributes above were used as direct inputs to the k-means clustering. The other 10 are either identifiers, intermediate calculation steps, pre-existing labels kept aside for validation rather than input, or attributes deliberately excluded because they would have distorted the clustering (see elasticity, Section 3 below).

2. Importance Ranking — Attributes Used in Clustering

The 10 attributes below were the actual inputs to the k-means clustering. Each was tested with an ANOVA F-test after clustering: this checks whether that attribute's average value genuinely differs across the 4 resulting clusters, or whether any apparent difference could just be random chance. A higher F-statistic means a stronger, more reliable differentiator between clusters. A result is generally considered statistically significant when $p < 0.05$.

Rank	Attribute	Source File	F-statistic	Significant? (p < 0.05)	What it captures
1	**stockout rate**	inventory.csv	354	Yes — strongest signal	How often the SKU ran out of stock. The single most dominant feature by a wide margin.
2	**lead time std days**	network_lanes.csv + plant_dc_master.csv	278	Yes	How unpredictable replenishment timing is.
3	**lead time mean days**	network_lanes.csv + plant_dc_master.csv	266	Yes	How long replenishment typically takes.
4	**list price usd**	sku_master.csv	260	Yes	The SKU's catalog price.
5	**revenue total**	orders.csv	79	Yes	Total dollars the SKU generated.
6	**demand cv**	demand_forecast.csv	56	Yes	How volatile weekly demand is.
7	**margin pct**	sku_master.csv	3	Yes — borderline	Profit margin %. A real but weak differentiator.
—	**supply adherence std**	production.csv	n/a	No	Production consistency — does not meaningfully differ across clusters.
—	**shelf life days**	sku_master.csv	n/a	No	Days until expiry — does not meaningfully differ across clusters.
—	**forecast mape**	demand_forecast.csv	n/a	No	Forecast accuracy — does not meaningfully differ across clusters.

Key takeaway: the segmentation is fundamentally driven by supply-side risk — stockout rate and lead time together form the top 3 signals — more than by revenue or demand volatility, which rank 5th and 6th. Production consistency, shelf life, and forecast accuracy (highlighted in red) do not meaningfully differ between the 4 clusters we formed.

3. Note — Why Price Elasticity Was Excluded From Clustering

Elasticity is calculated once per product category (5 categories total), not per individual SKU — an earlier per-SKU approach produced unstable, sometimes nonsensical results (including one case implying demand rises with price) because individual SKUs had too little price movement to fit a reliable regression. Pooling all SKUs within a category, with SKU and week fixed effects, gives a far more stable estimate.

However, this means elasticity only has 5 distinct values across all 500 SKUs. A feature with just 5 possible values dominates k-means distance calculations and would effectively turn the clustering into "which category is this SKU in" rather than finding genuine cross-category behavioral patterns. For this reason, elasticity is computed and retained in the output for reference and profiling, but was deliberately excluded from the clustering inputs themselves.